BIJOY BHASKAR LALL

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EDUCATION

Birla Institute of Technology and Science (BITS) Pilani, Pilani, India **Nov 2020 – May 2024**
Secured a Bachelor of Engineering in Mechanical Engineering with *Distinction* **(CGPA: 9.29/10)**
Merit Scholarship for being in the top 2% of the batch

PROFESSIONAL EXPERIENCE

Mercedes-Benz Research and Development India (MBRDI), Bengaluru | *CAE Analyst* **Oct 2024 – Present**

- Developed factory-planning digital twins using NVIDIA Omniverse for 3 carlines and onboarded 1,000+ planners, generating an estimated 30% reduction in annual planner-hours through reduced reliance on physical factory trials.
- Performed automotive interior component simulations using ANSA and ABAQUS across 10+ load cases and 30+ seat hardware tests to support design evaluation and development, seeing through 3 carlines to production kick-off.
- Built a prediction model to quantify perceived occupant comfort from seat simulation and sensor data, saving an estimated 600 hours of simulation time and €30,000 annually due to reduced simulation and software license requirements.
- Developed workflows to automate simulation model pre-processing, reducing validation time by 20% per carline project.
- Inventor of 4 patents (patent pending) to prevent serious injuries in vehicle crash scenarios.

Mercedes-Benz Research and Development India (MBRDI), Bengaluru | *Student Intern* **Jul 2023 – Dec 2023**

- Conducted a data-driven study of seat design and occupant comfort using Python and ABAQUS, analysing 500+ historical simulation data points with XGBoost to develop a prediction model projected to save €70,000 annually in software costs.
- Served as Master of Ceremonies of 2 company-level conferences, attended by 200+ delegates.
- Coordinated weekly managerial meetings for 80+ colleagues covering technical and leadership updates.

Birla Institute of Technology and Science Pilani, Pilani, India | *Teaching Assistant – Theory of Machines* **Jan 2023 – May 2023**

- Taught classes of 200+ students on using AutoCAD software to solve kinematic analysis problems involving link-chain mechanisms; conducted doubt-solving sessions.
- Assisted in the creation of exam questions for 5+ test components as well as in invigilation and paper correction.

Indira Gandhi Centre for Atomic Research (IGCAR), Kalpakkam, India | *Research Intern* **Jun 2022 – Jul 2022**

- Developed a kinematic model of the robot manipulator arm at the facility using the Denavit-Hartenberg approach, MSC ADAMS and AutoCAD, with an estimated saving of over ₹1 lakh in potential prototype cost avoidance.
- Presented the report to a panel of distinguished scientists at IGCAR, enabling its use in future simulation and control studies.

ACADEMIC PROJECTS

Kinematics of Humanoids for Bipedal Locomotion

- Developed a mathematical model of a bipedal humanoid robot using the Denavit-Hartenberg approach.
- Validated the forward and inverse kinematics of the model across 6 degrees of freedom using the MATLAB Robotics Toolbox.
- Used MSC ADAMS to digitally validate the model, supporting the development of robots for hazardous environments.

Vibrational Analysis of Ear Ossicles

- Established a correlation between the resonance frequency of the ear structure and changes in its density due to disease.
- Modelled the ear structure using Fusion 360, performed modal analysis in ANSYS across different material properties, and used exponential curve fitting to quantify the relationship, thus establishing a method for non-invasive disease diagnosis.

Design of Sustainable Campus Utility Solutions

- Designed sustainable, 3D-printable replacements for damaged campus utilities, developing a three-hook door hanger and a water tap aerator using FreeCAD; digitally simulated and validated the designs using CalculiX and OpenFOAM.
- Incorporated screw- and adhesive-free mounting and evaluated biodegradable material options to improve sustainability.
- Assessed manufacturability, with estimated 34% and 47% lower costs for the hanger and aerator respectively.

TECHNICAL SKILLS & CERTIFICATIONS

- **CAE, CAD & Simulation:** ANSA, ABAQUS, ANSYS, LS-DYNA, OpenFOAM, METAPOST, MSC ADAMS, Siemens NX, Fusion 360
- **Programming & Data Analytics:** Python, MATLAB, Excel for Data Analytics
- **Certifications:** AI Foundations: Machine Learning, Python for Data Science and Machine Learning

EXTRACURRICULAR ACTIVITIES

- Topic Driver, Innovation Cluster, MBRDI (May 2025 - Present): Lead ideation and filing for 10+ patents.
- Member of the BITS Pilani Cricket Team (March 2022 - May 2023): Organized annual inter-college tournaments and coordinated the preparation of facilities; captained the school team and earned awards at state-level tournaments.
- Founder & Vice-President, Model United Nations (MUN) Club, Modern School (2019 - 2020): Won national-level accolades.
- I.P. Gupta Scholarship for Best Male Scholar (2019): Awarded to students ranking in the top 1% of the science stream.
- Social Venture: Sardar Bhagwant Singh award for excellence in research work (2018): Outreach program to educate students on the constructive use of technology. Held classes and sessions at school and undertook door-to-door community outreach.
- Volunteer, WWF-India (2017 - 2018): Conducted wildlife workshops and contributed research for WWF-India publications.